

$$Q_1) a) 1280 \times 720 = 921,600 \text{ pixels}$$

$$bpf = 29491200$$

$$29491200 = 921600 \times \lambda$$

$$\lambda = \frac{29491200}{921600}$$

$$\lambda = 32 \text{ bits/pixel}$$

$$b) 1 \text{ byte} = 8 \text{ bits}$$

$$\text{bytes per frame} = \frac{29491200}{8}$$

$$= 3686400$$

$$= 3.69 \text{ MB}$$

$$c) 70 \text{ MB/s} \quad 30 \text{ fps}$$

$$1 \text{ byte} = 8 \text{ bits}$$

$$70 \times 8 = 560 \text{ Mb/s}$$

$$Q_2) A1 : Q[1.5, 3], K[1.1, 2.2], V[0.2, 0.3]$$

$$\text{Governance: } Q[2, 4], K[1, 3], V[2, 0.3]$$

$$\text{Emerging: } Q[2.2, 4.4], K[1, 3], V[2.2, 0.3]$$

$$\text{Course: } Q[2.4, 4.4], K[1.1, 3], V[2.2, 0.3]$$

$$dk = 2$$

$$\text{Attention} = \left(\frac{QK^T}{\sqrt{dk}} \right) V$$

$$Q = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 4.4 \\ 2.4 & 4.4 \end{bmatrix} \quad K = \begin{bmatrix} 1.1 & 1 & 1 & 1.1 \\ 2.2 & 3 & 3 & 3 \end{bmatrix} K^T$$

$$QK^T \Rightarrow \begin{bmatrix} (1.5)(1.1) + (3)(2.2) & (1.5)(1) + (3)(3) + (1.5)(1) + (3)(3) & (1.5)(1.1) + (3)(3) \\ (2)(1.1) + 4(2.2) & 2(1) + 4(3) & 2(1.1) + 4(3) \\ 2.2(1.1) + 4.4(2.2) & 2.2(1) + 4.4(3) & 2.2(1.1) + 4.4(3) \\ 2.4(1.1) + 4.4(2.2) & 2.4(1) + 4.4(3) & 2.4(1.1) + 4.4(3) \end{bmatrix}$$

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$$\frac{QK^T}{\sqrt{2}} \Rightarrow \begin{bmatrix} 8.25 & 10.5 & 10.5 & 10.65 \\ 11 & 14 & 14 & 14.2 \\ 12.1 & 15.4 & 15.4 & 15.62 \\ 0.0300 & 0.3046 & 0.3046 & 0.3609 \end{bmatrix}$$

$$\frac{QK^T}{\sqrt{2}} \Rightarrow \begin{bmatrix} 5.8336 & 7.4246 & 7.4246 & 7.5307 \\ 7.7782 & 9.8995 & 9.8995 & 10.0409 \\ 8.5560 & 10.8894 & 10.8894 & 11.0450 \\ 8.7116 & 11.0309 & 11.0309 & 11.2086 \end{bmatrix}$$

normalisation via
rowwise softmax
downward $\rightarrow$

$$\Rightarrow \begin{bmatrix} 0.0614 & 0.3016 & 0.3016 & 0.3354 \\ 0.0366 & 0.3056 & 0.3056 & 0.3521 \\ 0.0297 & 0.3063 & 0.3063 & 0.3578 \\ 0.0300 & 0.3046 & 0.3046 & 0.3609 \end{bmatrix} \times \begin{bmatrix} 0.2 & 0.3 \\ 2 & 0.3 \\ 2.2 & 0.3 \\ 2.2 & 0.3 \end{bmatrix}$$

$$\left(\frac{QK^T}{\sqrt{2}}\right)^v = \begin{bmatrix} 2.0168 & 0.3000 \\ 2.0656 & 0.3000 \\ 2.0794 & 0.3000 \\ 2.0792 & 0.3000 \end{bmatrix}$$



Q3)

Wind	Humidity	Rain
Weak	high	yes
Strong	high	yes
Weak	low	no
Weak	low	yes
Strong	low	no

1) Laplace smoothing (Naive Bayes):

$$P(\text{Rain}) = \frac{\text{Yes}}{3} = 0.6, \quad P(\text{No Rain}) = \frac{\text{NO}}{2} = 0.4$$

$$P(x=v|c) = \frac{\text{count}(x=v, c) + 1}{\text{count}(c) + k}; \quad k=2 \text{ (wind \& humidity)}$$

$$P(\text{yes}|\text{rain}) \Rightarrow \text{Wind: } P(\text{weak}|\text{yes}) = \frac{2+1}{3+2} = \frac{3}{5} = 0.6$$

$$P(\text{strong}|\text{yes}) = \frac{1+1}{3+2} = \frac{2}{5} = 0.4$$

$$\text{Humidity: } P(\text{high}|\text{yes}) = \frac{2+1}{5} = \frac{3}{5} = 0.6$$

$$P(\text{low}|\text{yes}) = \frac{1+1}{5} = \frac{2}{5} = 0.4$$

$$P(\text{NO}|\text{rain}) \Rightarrow \text{Wind: } P(\text{weak}|\text{no}) = \frac{1+1}{2+2} = \frac{2}{4} = 0.5$$

$$P(\text{strong}|\text{no}) = \frac{1+1}{2+2} = \frac{2}{4} = 0.5$$

$$\text{Humidity: } P(\text{high}|\text{no}) = \frac{0+1}{2+2} = \frac{1}{2} = 0.25$$

$$P(\text{low}|\text{no}) = \frac{2+1}{2+2} = \frac{3}{4} = 0.75$$

a. weak wind + high humidity:

$$\text{rain} = \text{yes} : P(\text{yes}) \times P(\text{weak}|\text{yes}) \times P(\text{high}|\text{yes})$$

$$0.6 \times 0.6 \times 0.6 = 0.216$$

$$\text{rain} = \text{no} : P(\text{no}) \times P(\text{weak}|\text{no}) \times P(\text{high}|\text{no})$$

$$0.4 \times 0.5 \times 0.25 = 0.05$$

$$\text{normalise yes} : \frac{0.216}{0.216 + 0.05} = 0.8120$$

$$\text{normalise no} : \frac{0.05}{0.216 + 0.05} = 0.1880$$

Rain = yes ; don't deliver package

b. strong wind + high humidity:

$$\text{rain} = \text{yes} : P(\text{yes}) \times P(\text{strong}|\text{yes}) \times P(\text{high}|\text{yes})$$

$$0.6 \times 0.4 \times 0.6 = 0.144$$

$$\text{rain} = \text{no} : P(\text{no}) \times P(\text{strong}|\text{no}) \times P(\text{high}|\text{no})$$

$$0.4 \times 0.5 \times 0.25 = 0.05$$

$$\text{yes} = \frac{0.144}{0.144 + 0.05} = 0.7423$$

$$\text{no} = \frac{0.05}{0.144 + 0.05} = 0.2577$$

Rain = yes ; don't deliver package

c. weak wind + low humidity:

$$\text{rain} = \text{yes} = 0.6 \times 0.6 \times 0.4 = 0.144$$

$$\text{rain} = \text{no} = 0.4 \times 0.5 \times 0.75 = 0.15$$

$$\text{yes} = \frac{0.144}{0.144 + 0.15} = 0.4898$$

$$\text{no} = \frac{0.15}{0.144 + 0.15} = 0.5102$$

Drone should
Rain = no ; deliver package

d. strong wind + low humidity:

$$\text{rain (yes)} = 0.6 \times 0.4 \times 0.4 = 0.096$$

$$\text{rain (no)} = 0.4 \times 0.5 \times 0.75 = 0.15$$

$$\text{yes} = \frac{0.096}{0.096 + 0.15} = 0.3902$$

$$\text{no} = \frac{0.15}{0.096 + 0.15} = 0.6092$$

Rain = no; Drone should deliver package

2) Entropy & Info Gain:

$$H(S) = -\sum p_i \log_2 p_i$$

$$H(\text{rain}) = -\frac{3}{5} \log_2 \left(\frac{3}{5} \right) - \frac{2}{5} \log_2 \left(\frac{2}{5} \right) = -0.44218 - 0.52877 = 0.97095$$

wind = strong; yes = 1, no = 1

$$H(\text{strong}) = -\frac{1}{2} \log_2 \left(\frac{1}{2} \right) - \frac{1}{2} \log_2 \left(\frac{1}{2} \right) = 1$$

wind = weak; yes = 2, no = 1

$$H(\text{weak}) = -\frac{2}{3} \log_2 \left(\frac{2}{3} \right) - \frac{1}{3} \log_2 \left(\frac{1}{3} \right) = 0.91830$$

$$\text{weighted entropy} = H(\text{rain}|\text{wind}) = \frac{3}{5} (0.91830) + \frac{2}{5} (1)$$

$$= 0.95098$$

$$\text{Information Gain} = H(\text{rain}) - H(\text{rain}|\text{wind})$$

$$= 0.97095 - 0.95098$$

$$\boxed{= 0.01997}$$

humidity = high; yes = 2, no = 0

$$H(\text{high}) = -\frac{2}{2} \log_2 \left(\frac{2}{2} \right) - \frac{0}{2} \log_2 \left(\frac{0}{2} \right) = 0$$

humidity = low; yes = 1, no = 2

$$H(\text{low}) = -\frac{1}{3} \log_2 \left(\frac{1}{3} \right) - \frac{2}{3} \log_2 \left(\frac{2}{3} \right) = 0.91830$$

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$$\text{weighted entropy} = h(\text{rain} / \text{nonidling}) = \frac{2}{5}(0) + \frac{3}{5}(0.91830) \\ = 0.55098$$

$$\text{Information Gain} = 0.97095 - 0.55098 \\ = \boxed{0.41997}$$

∴ since information gain of nonidling is higher it is the ~~best~~ ^{root} to be used for further calculation.

Q3) Logistic:

wind: weak - 0
strong - 1humidity: low - 0
high - 1rain: no - 0
yes - 1wind (x_1)humidity (x_2)rain (y)

0

1

1

1

1

1

0

0

0

0

0

1

1

0

0

$$z = w_1x_1 + w_2x_2 + b$$

$$P(\sigma(z)) = \frac{1}{1+e^{-z}} ; \lambda = 0.1$$

 $P \geq 0.5 \Rightarrow \text{yes}$
 $P < 0.5 \Rightarrow \text{no}$

model parameters:

$$z = 0.048624 - 0.539200x_1 + 1.616261x_2$$

• weak wind + high humidity:

$$x_1 = 0, x_2 = 1$$

$$z = 0.048624 - 0.539200(0) + 1.616261(1) \\ = 1.664885$$

$$P(\text{yes}) = \frac{1}{1+e^{-1.664885}} = 0.8409$$

 $0.8409 > 0.5 ; \text{rain} = \text{yes} \therefore \text{drone shouldn't deliver}$

• strong wind + high humidity:

$$x_1 = 1, x_2 = 1$$

$$z = 0.048624 - 0.539200(1) + 1.616261(1)$$

$$z = 1.125685$$

$$P(\text{yes}) = \frac{1}{1+e^{-1.125685}} = 0.7550$$

 $0.7550 > 0.5 ; \text{rain} = \text{yes} \therefore \text{drone should not deliver}$

- weak wind + low humidity:

$$x_1 = 0, x_2 = 0$$

$$z = 0.048624 - 0.539200(0) + 1.616261(0)$$

$$z = 0.048624$$

$$P(\text{yes}) = \frac{1}{1 + e^{-0.048624}} = 0.5122$$

$0.5122 > 0.5$; rain = yes $\therefore$ drone shouldn't deliver

- Strong wind + low humidity:

$$x_1 = 1, x_2 = 0$$

$$z = 0.048624 - 0.539200(1) + 1.616261(0)$$

$$z = -0.490576$$

$$P(\text{yes}) = \frac{1}{1 + e^{+0.490576}} = 0.3798$$

$0.3798 < 0.5$; rain = no $\therefore$ drone should deliver package

4) - Naive Bayes is a probabilistic classifier which works well with small dataset & features. However it assumes that these features like wind & humidity in this case are independent. The result of this algo. showed humidity strongly affects rain.

- Entropy & Info. gain are decision tree ~~steps~~ ~~to~~ which classify which feature is most useful for classification. In this algo, humidity was again the feature which had higher info gain so it was better for predicting rain.

- Logistic Regression predicts probability using weighted combination of features & sigmoid function.

Overall, humidity is the strongest predictor.

Q5) A = Rice Field, B = Cotton Field; H = healthy, I = infected

- \$2 credit for healthy field at each timestamp

- if infected spray, if healthy move

- field stays healthy for 10 timestamps.

• independent with equal probability

$$1) P(A=I, B=I) = P(A=I) \times P(B=I)$$

$$= 0.5 \times 0.5$$

$$= 0.25$$

$$= 25\%$$

probability of both fields being infected.

2) A & B both infected initially. Drone is currently monitoring A.

Table:

Time	Location	Percept	Action	After Action State	Credit
0	A	[A, I]	Spray A	[A=H], [B=I]	\$0
1	A	[A, H]	move to B	[A=H], [B=I]	\$2
2	B	[B, I]	spray B	[A=H], [B=H]	\$0
3	B	[B, H]	move to A	[A=H], [B=H]	\$2

Total credit = \$4

Q7) 1) $A = (6, -1, \sqrt{9})$ $B = (4, x, -2)$; for what value of x is $B \perp A$

$$A \cdot B = 0$$

$$(6)(4) + (-1)(x) + (\sqrt{9})(-2) = 0$$

$$24 - x - 6 = 0$$

$$\boxed{x = 18}$$

2) Dot Product ; multiplies elements ~~wise~~ ^{correspondingly} and then adds them together like in part 1. $[A \cdot B = a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n]$ resulting in a scalar.

Hadamard Product ; multiplies element wise resulting in a vector.

$$A \odot B = a_1b_1, a_2b_2, a_3b_3$$

$$3) u = (1, 2) \quad v = (2, 2)$$

$$u \cdot v = (1)(2) + (2)(2) \quad - \text{scalar projection}$$

$$= 6$$

$$\text{vector projection} = \frac{u \cdot v}{\|v\|^2} v$$

u projection on $(\frac{3}{2}, \frac{3}{2})$ as it follows the line $y=x$

$$\|u_{\text{proj}}\| = \sqrt{(\frac{3}{2})^2 + (\frac{3}{2})^2} = \frac{3\sqrt{2}}{2}$$

$$\|v\| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

$$\text{proj } u = \frac{u \cdot v}{\|v\|}$$

$$u \cdot v = \frac{3\sqrt{2}}{2} \times 2\sqrt{2}$$

$$\boxed{u \cdot v = 6} \rightarrow \text{proved}$$

$$Q_8) S(t) = A(t) + B(t) + C(t) + D(t)$$

where :-

$$A(t) = 0.2t + t^2$$

$$B(t) = \sin(t) + \cos(t)$$

$$C(t) = t^5 + t^6 + \sin(t)$$

$$D(t) = t^5 + t^8 + \sin(t) + \cos(t)$$

~~C(t)~~

$$t = 1 :$$

$$A(1) = 0.2 + 1^2 = 1.2$$

$$B(1) = \sin(1) + \cos(1) = 1.0173$$

$$C(1) = 1^5 + 1^6 + \sin(1) = 2.0174$$

$$D(1) = 1^5 + 1^8 + \sin(1) + \cos(1) = 3.0173$$

$$S(1) = 7.252$$

$$t = 4 :$$

$$A(4) = 16.8$$

$$B(4) = 1.067$$

$$C(4) = 5120.07$$

$$D(4) = 66561.067$$

$$S(4) = 71699.004$$

$$t = 2 :$$

$$A(2) = 4.4$$

$$B(2) = 1.0342$$

$$C(2) = 96.0348$$

$$D(2) = 289.0342$$

$$S(2) = 390.5034$$

$$t = 3 :$$

$$A(3) = 9.6$$

$$B(3) = 1.051$$

$$C(3) = 972.052$$

$$D(3) = 6805.051$$

$$S(3) = 7787.754$$

